

UC1: Database Table Creation and Population

A.

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
Table Name	String	Must be one of: users, courses, enrollments, class_sessions, attendance_records	Recognized table name matching schema definition	Unrecognized or misspelled table name	Empty string or NULL
SQL Statement	String (DDL/DML)	Valid SQL for INSERT, SELECT, TRUNCATE, or DDL; must reference existing tables/columns	Well-formed SQL matching schema	SQL referencing non-existent columns or tables	Malformed SQL syntax
Field Value	Varies (String / Boolean / Datetime)	Must satisfy column constraints: NOT NULL, UNIQUE, FK reference, CHECK rules	Value satisfying all column constraints	NULL where NOT NULL required; duplicate where UNIQUE required	Control characters, oversized strings
Foreign Key Reference	String / Integer	Must reference an existing row in the parent table (e.g. instructor_netid → users.netid)	netid/course_id that exists in the parent table	netid/course_id that does not exist	DB unavailable during lookup
Datetime / Interval	Datetime	start_time must be before end_time; attendance_window_end must be >= start_time	start < end with valid timestamps	end_time < start_time (violates CHECK constraint)	Server clock failure or NULL timestamp

B.

Test #	Test Scenario	Input	Action
1	Tables exist after schema applied	Schema SQL executed against test DB	Query information_schema.tables for each expected table
2	Valid user insert	All required fields provided and valid	INSERT into users with valid netid, name, email, password_hash, is_instructor, created_at

3	Duplicate primary key	Two INSERTs with the same netid	Second INSERT into users with duplicate netid
4	NOT NULL violation	name field set to NULL	INSERT into users with NULL name
5	Unique email constraint	Two users with identical email	Second INSERT using already-used email address
6	Foreign key violation	instructor_netid references non-existent user	INSERT into courses with instructor_netid = 'nonexist'
7	CHECK constraint on session times	end_time set 1 hour before start_time	INSERT into class_sessions where end_time < start_time

C.

Test #	Concrete Input Values	Action	Expected Outcome
1	table_name in ['users','courses','enrollments','class_sessions','attendance_records']	SELECT EXISTS FROM information_schema.tables WHERE table_name = %s	table_exists = True for all 5 tables
2	netid='ab1234', name='John Doe', email='john@example.com', password_hash='hash', is_instructor=False	INSERT INTO users ... VALUES ('ab1234','John Doe','john@example.com','hash',False,NOW())	fetchone() returns non-None row
3	First: netid='ab1234'. Second: netid='ab1234' (duplicate)	Two consecutive INSERTs with same netid='ab1234'	psycopg.errors.UniqueViolation raised on second INSERT
4	netid='cd5678', name=None, email='test@test.com', password_hash='hash', is_instructor=False	INSERT INTO users with name=None (%s bound to Python None)	psycopg.errors.NotNullViolation raised
5	First: netid='ab1234', email='dup@email.com'. Second: netid='cd5678', email='dup@email.com'	Two INSERTs sharing email='dup@email.com'	psycopg.errors.UniqueViolation raised on second INSERT

6	course_id='CS101', course_name='Intro to CS', instructor_netid='nonexistent', enrollment_code='ABCDEF GH'	INSERT INTO courses with instructor_netid='nonexistent' (no matching users row)	psycopg.errors.ForeignKeyViolation raised
7	course_id='CS101', start_time=now, end_time=now - timedelta(hours=1), attendance_window_end=now	INSERT INTO class_sessions with end_time < start_time	psycopg.errors.CheckViolation raised

D.

```

PS C:\Users\sh100\code\Group-06> pytest tests/test_db.py -v
===== test session starts =====
platform win32 -- Python 3.12.3, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\sh100\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\sh100\code\Group-06
plugins: anyio-4.11.0
collected 7 items

tests/test_db.py::test_tables_created PASSED [ 14%]
tests/test_db.py::test_valid_insert_user PASSED [ 28%]
tests/test_db.py::test_duplicate_primary_key PASSED [ 42%]
tests/test_db.py::test_not_null_violation PASSED [ 57%]
tests/test_db.py::test_unique_email PASSED [ 71%]
tests/test_db.py::test_foreign_key_violation PASSED [ 85%]
tests/test_db.py::test_check_constraint PASSED [100%]

===== 7 passed in 1.16s =====

```

UC2: Attendance Export as .csv

E.

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
Session netid	Session	Must exist in Flask Session	Recognized netid present	Missing netid	Session store failure

Session is_instructor	Session flag	Boolean that must be true	True	False	Missing flag
course_id	String	Non-empty string with course identifier	Recognized course	“”, None	Extremely large or malformed string
Attendance rows	Database query result (list of dicts)	List of attendance records	List with valid dict rows	Empty list	Database exception
Attendance row field	Varies	Value indicating data for that row	Value matching row expectation (eg: Valid netid for netid row)	Missing	Corrupt record

F.

Test #	Test Scenario	Input	Action
1	Valid export	Session: netid=proftest, is_instructor=True; course_id=CS3354.001; Valid course and exported file	Call /attendance/export/course_id=CS3354.001 -> expect 200 response
2	Not authenticated	No session	Call endpoint -> expect 401 not authenticated
3	Student access denied	Session: valid netid, is_instructor = False	Call endpoint -> expect 403 access denied
4	Missing course_id	Session: valid instructor; no course_id param	Call attendance/export -> Expect 400 error missing course id
5	Course not found	Session: valid instructor; course_id=CS3354.001; repository.get_course -> None	Call endpoint -> expect 404 course not found
6	Empty list	Session: valid instructor, valid course; get_course_attendance -> []	Call endpoint -> expect 200 CSV with only header

7	Repository exception (course lookup failure)	Session: valid instructor; get_course raises exception	Call endpoint -> expect 500 Internal Server Error
8	Malformed attendance row (missing field)	Session: Valid instructor; attendance row missing "name"	Call endpoint -> expect 500 internal server error

G.

Test #	Concrete Input Values	Action	Expected Outcome
1	Session: netid=proftest, is_instructor=True; course_id=CS3354.001; course exists; attendance has valid rows	GET /attendance/export?course_id=CS3354.001	200 OK, CSV returned correctly
2	No session data; course_id=CS3354.001	GET /attendance/export?course_id=CS3354.001	401 {"error": "not authenticated"}
3	Session: netid=dal123456, is_instructor=False; course_id=CS3354.001	GET /attendance/export?course_id=CS3354.001	403 {"error": "access denied"}
4	Session: netid=proftest, is_instructor=True; no course_id provided	GET /attendance/export	400 {"error": "missing course_id"}
5	Session: netid=proftest, is_instructor=True; course_id=CS3354.001; get_course → None	GET /attendance/export?course_id=CS3354.001	404 {"error": "course not found"}
6	Session: valid instructor; course_id=CS3354.001; get_course_attendance → []	GET /attendance/export?course_id=CS3354.001	200 OK, CSV with only header row
7	Session: valid instructor; course_id=CS3354.001	GET /attendance/export?course_id=CS3354.001	500 Internal Server Error

	1; get_course → raises Exception		
8	Session: valid instructor; attendance row missing name field	GET /attendance/export?course_id=CS3354.001	500 Internal Server Error (KeyError during CSV generation)

H.

The screenshot shows the PyCharm IDE interface. The top toolbar includes buttons for Run, Test, and other development tools. The left sidebar shows the project structure with folders like .venv, db-schema-seeding, and Phase 4 - Static Files. The main editor displays the test_export.py file with the following code:

```

17 def login_student(client):
18     with client.session_transaction() as sess:
19         sess['netid'] = 'dal123456'
20         sess['is_instructor'] = False
21         sess['password'] = 'password1234*'
22
23 def login_instructor(client):

```

The bottom panel shows the test results for the run of test_export.py. The results indicate that 8 tests passed out of 8 tests total, taking 22 ms. The test cases and their progress are as follows:

Test Case	Progress
tests/test_export.py::test_export_attendance_valid PASSED	[12%]
tests/test_export.py::test_export_attendance_not_authenticated PASSED	[25%]
tests/test_export.py::test_export_attendance_student_access_denied PASSED	[37%]
tests/test_export.py::test_export_attendance_instructor_access_denied PASSED	[50%]
tests/test_export.py::test_export_attendance_course_not_found PASSED	[62%]
tests/test_export.py::test_export_attendance_empty PASSED	[75%]
tests/test_export.py::test_export_attendance_repository_exception PASSED	[87%]
tests/test_export.py::test_export_attendance_malformed_row PASSED	[100%]

The test session summary shows: 8 passed in 0.52s. The process finished with exit code 0.